



UNIVERSITY
OF TRENTO - Italy

Dipartimento di Ingegneria e Scienza dell'Informazione

Know
dive



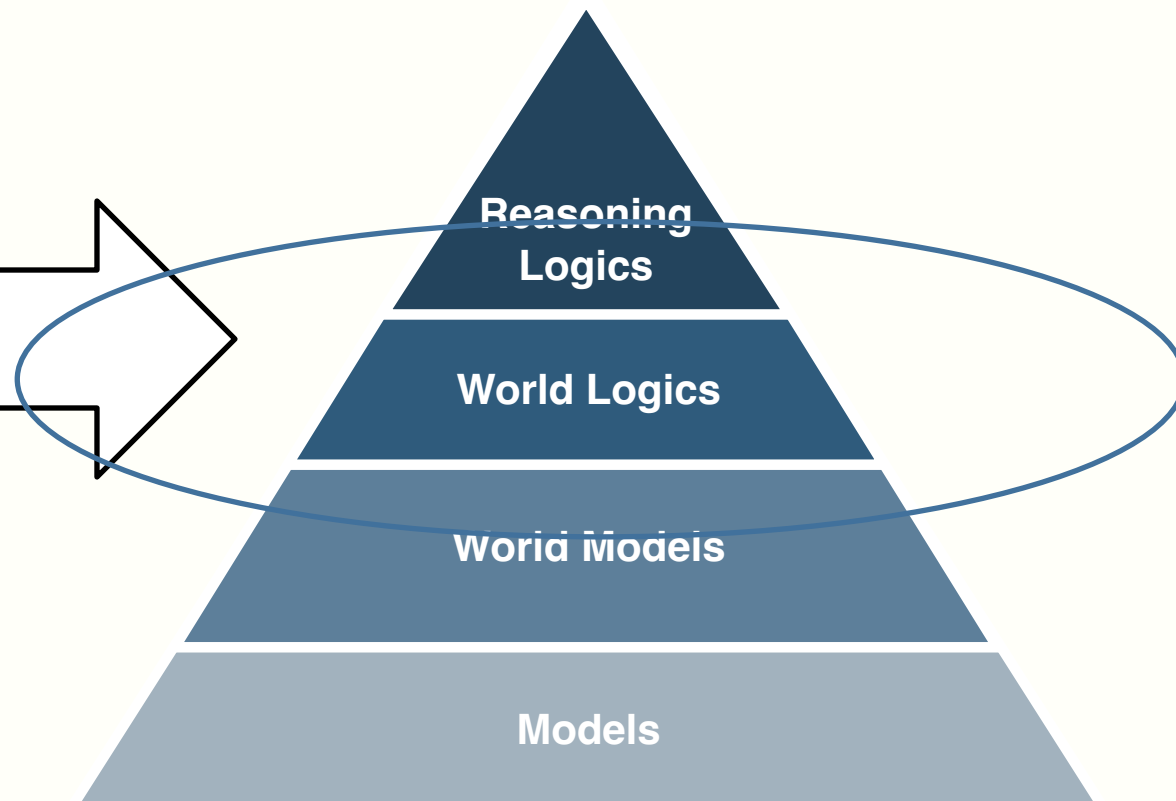
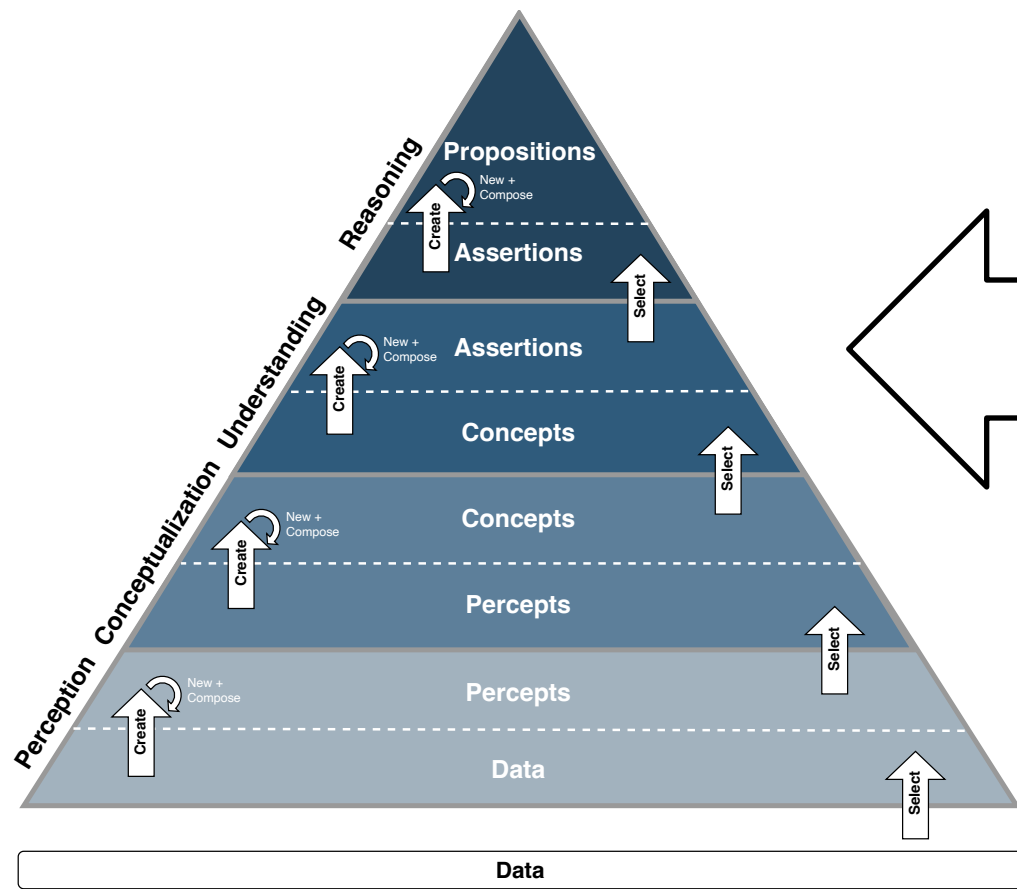
World logics

(HP2T)

LoE – The Logic of entities

- **Introduction**
- World Entailment and Logics
- Logical Entailment
- Properties of Logical Entailment
- Properties of World Logics
- World Logics diversity
- Key notions

The knowledge Pyramid and the Logics pyramid (recall)



Maximal theories (recall)

Definition (Maximal theory). A theory is **maximal** with respect to M if it is complete with respect to M. **Not maximal** or **incomplete**, otherwise.

Observation 1 (Intended model, incomplete theory). The intended model has been informally defined as the model one has in mind when writing the theory. But, because of the missing requirement on completeness, an incomplete theory has multiple models, only one of which is the intended model.

Examples (Intended model, incomplete theory).

- An ER model which does not specify the relation between students and professors
- A Table describing people's job which does not specify the people's names
- The description of a photo of a dog and a person which does not have etypes "dog" and "person"

Observation 2 (Intended model, incomplete theory). The fact that the intended model is partially specified implies that the theory cannot describe all that is relevant to the problem to solve. This is the cause of hallucinations in LLMs and design mistakes in SwEng.

Canonical models (recall)

Definition (Canonical model). Given a theory, the **canonical model** for that theory is the model for which T_a is maximal.

Observation 1 (Maximal theory). Given a world model there may be multiple maximal theories, that is theories which contain one or more assertions for each fact in the intended model. This is because of synonyms.

Observation 2 (Maximal theory). With no synonyms, there is one and only one maximal theory.

Observation 3 (Canonical model, intended model). When the theory is maximal, canonical model and intended model coincide. When this is not the case, we assume that the theory, and therefore, the canonical model are incomplete and do not fully describe what is the case.

Observation 4 (Incompleteness of the canonical model). Language structures how we think about the world, but it is the world, that is, the intended model (as a mental analogic representation or as an intended model), which defines what is the case. Which is why we may have possibly incomplete /incorrect world model linguistic descriptions

Observation 5 (Canonical world representation). A world representation is **canonical** when M is the **canonical model** of T_a (that is, T_a is **maximal** for M).

Canonical World models (recall)

Definition (Canonical world model). A world model

$$W = \langle L_a, D, I_a \rangle$$

is a **canonical world model** if

$$D = \{f\} = \{I_a(a) = f, \text{ for all } a \in L_a\}$$

That is:

$$W = \langle L_a, I_a(L_a), I_a \rangle = \langle L_a, I_a \rangle$$

Canonical World representations (recall)

Theorem (Canonical world representation). Given a canonical world model

$$W = \langle L_a, I_a \rangle$$

then

$$R = \langle T_a \rangle$$

is a **canonical world representation**, with $T_a = \{a\} \subseteq L_a$, and

$$M = \{f\} = \{I_a(a) = f, \text{ for all } a \in T_a\} \subseteq D$$

where M and T_a are, respectively, a **model** of T_a and a **theory** of M in W .

Reminder (completeness). T_a is **complete** with respect to M , If, for all $a \in L_a$, $I_a(a) \in M$ then $a \in T_a$.



Semi-formal vs. (canonical) world models (recall)

Observation (Base and composed formal world models). We have provided 3 types of base world models further differentiated depending on (i) the level of generality of the root node and, in the case of etype graphs also, on (ii) the nesting of etypes with respect to the root concept.

Composed world models are obtained from base world model by making explicit the ISA and MG/LG hierarchies of concepts and etypes, respectively. As written above, differently from base world models, composed world models are complete representations of the relations holding among concepts, etypes and entities.

Observation (Semi-formal and formal world models). Semi-formal world models are ambiguous representations of formal base world models of various kinds. There are two possible types of mistakes:

- Ambiguity of the words used (words instead of concepts) and the syntax used to make assertions.
This problem is solved by canonical based world models.
- Incompleteness of the semi-formal world model (as it relates to the incompleteness of formal base models). This problem cannot be solved by world models (e.g., Wordnet as 100+ thousand nodes and millions of links). The unfolding of etypes into more specific etypes must be done dynamically and purpose driven. This problem is solved by **world logics**, via the **formalization of the reasoning mechanisms** underlying **unfolding** and **expansion**.

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World Logics

A World logic is

- A world model, also called the **reference world model**
- An operation which allows you to decide whether a certain assertion is a consequence of the reference world model. This is what we call **world entailment**.

Intuition (World Logic). A World Logic is a representation of the world which allows to reason about what is known of the reference world model, that is, **to deduce assertions which are implied by the known assertions**.

Observation (world entailment). Different logics are associated with different forms of world entailment.

Terminology (deduction, derivation, inference). The words **deduction**, **derivation**, and **inference** are sometimes used to mean the (reasoning) process by which new assertions are entailed from what is already known.

World entailment

Definition (World entailment) Let $W = \langle L_a, D, I_a \rangle$ be a world model. Let L_a be an assertional language. Then $\models_{L_a}$ is a **world entailment relation**

$$\models_{L_a} \subseteq D \times L_a$$

such that

$$M \models_{L_a} T_a$$

$M = \{f\} \subseteq D$ is a **model of** $T_a \subseteq L_a$. We say that M **entails** T_a . We write $M \models T_a$ instead of $M \models_{L_a} T_a$ when no confusion arises.

Intuition (World entailment). The model, our analogical representation of the world encodes all we know all the world. Most of the time only partially (incompletely). Reasoning, modeled as entailment, may help us to derive some facts thanks to some general reasoning rules, i.e., entailment rules, which model how the world is and evolves

World entailment – LoDE example

Entity types (LoD language):

- Church
- Building

Entities (LoE language):

- Duomo

Properties (LoD language):

- hasBellTower

Assertions (LoD Language):

- Church(Duomo)
- Church $\sqsubseteq$ Building $\sqcap$ hasBellTower

Assertions (LoDE language) derived via reasoning:

- Building(Duomo)
- hasBellTower(Duomo)

World Entailment - observations

Observation (World entailment relation). The entailment relation $\models$, as used in $M \models T$ allows to decide whether M is a model of T despite the non-maximality (incompleteness) of T , under the assumption of the correctness of T . This is not possible with world models which are (understood as) complete representations of the world.

Observation (World entailment relation). The entailment relation $\models$ is such that there is no longer a one-to-one mapping between the facts of the model and the assertions in the theory. Whether $M \models T$ must be computed via entailment, i.e., reasoning.

Intuition (World entailment relation). The world entailment relation allows to derive new assertions from what is already known (the model modelling the analogical memory). The key intuition is to extend the language of the reference world model, as used to express T , so that it is possible to express information that the very restricted syntax of world models does not allow to express.

Example (World entailment relation). The paradigmatic examples by which the language of world logics extends the language of world models is: (1) **Conjunction**, as in «a mother is a parent **and** a woman», (2) **Negation**, as in «a woman is **not** a man» and (3) **Disjunction**, as in «a person is a man **or** a woman»).

None of these three types of assertions can be made in any of the world models we have seen before.

World entailment – LoDE example

Entity types (LoD language):

- Church
- Building

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Assertions (LoD Language):

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Assertions (LoDE language) derived via reasoning:

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World Entailment - observations

Observation (World many-to-many entailment relation). World entailment is a many-to-many relation. That is, depending on the language used, which usually extends the world model language, a theory may have multiple models, and vice versa. In which case the entailment relation will hold for all models and only for them.

Example (World entailment many-to-many relation). A theory which contains the assertion «a person is a man **or** a woman» has a model where a person is a man and another model where a person is a woman.

World Logics

Proposition (World Logic). Let $W' = \langle L_a', D', I_a' \rangle$ be a World Model. Then a **World Logic** L_W for W is defined as

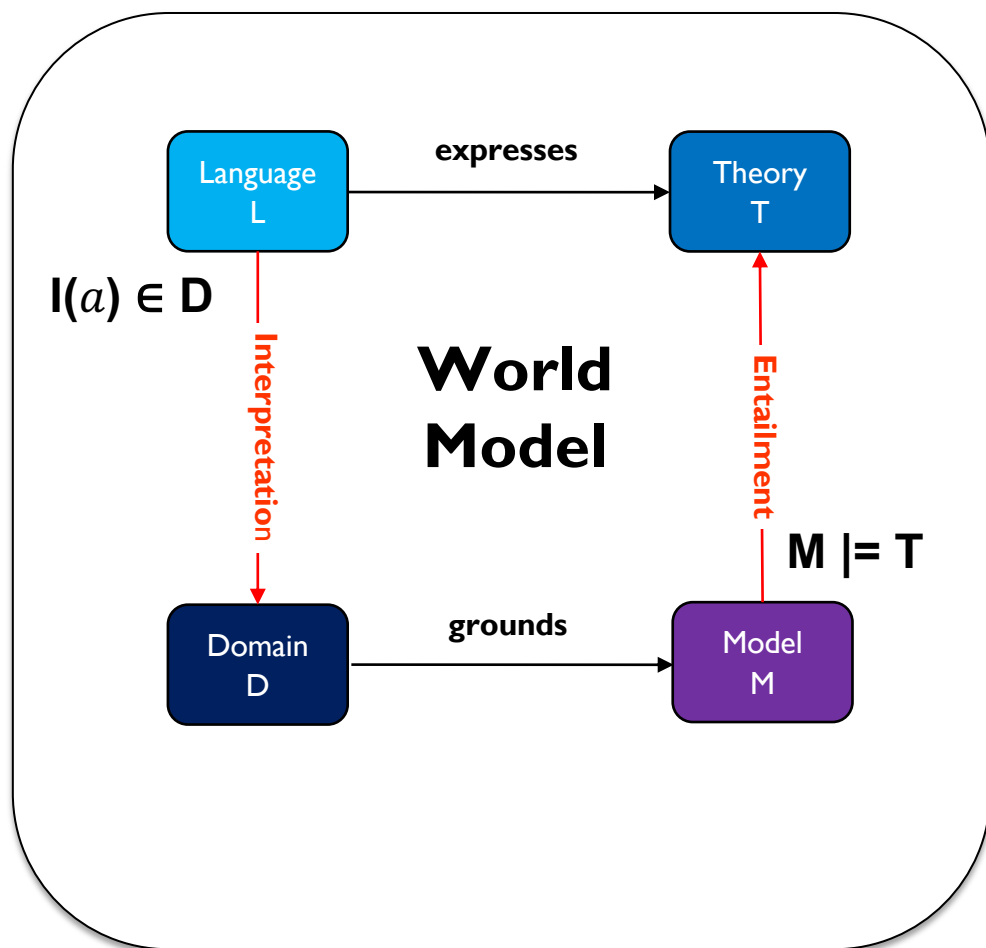
$$L_W = \langle W, |=_W \rangle$$

with $W = \langle L_a, D, I_a \rangle$ and $L_a' \subseteq L_a$, $D' \subseteq D$ and $I_a' \subseteq I_a$.

World Entailment vs interpretation – A comparison

- Interpretation is a function between a language and a domain: given an element of the language it returns the corresponding element of the domain of interpretation. It is used to “**construct**” **the domain of interpretation** (that is, our analogic representation of what we can potentially know of the world) starting from the language (that is, our linguistic representation of what we can potentially say about the world)
- World entailment is a relation between a model and a theory: given a set of assertions it assesses whether these assertions are actually a theory of the intended model. It is used to “**construct**” **the theory** (that is, our linguistic representation of what we know of the world) starting from (our analogical representation of what we know of the world)
- While both relating syntax and semantics, world entailment and interpretation function work in opposite directions: the interpretation function constructs the semantics, the entailment relation checks it for the truth of a set of input assertions.

World entailment and interpretation



Inside a world logic, world entailment and interpretation are related in the following way:

1. The interpretation function is exploited to **tell** a world logic the intended domain and then (by grounding, see later) the intended model via a set of assertions $a \in L$;
2. World entailment is exploited to **ask** a world logic whether a theory T is true in the intended model, as constructed, via the **tell** operation, by the interpretation function.

World models and world logics

$$\begin{array}{ccccc}
 a & \leftarrow \in & \text{---} & \mathcal{T}_a & \text{---} \subseteq \rightarrow \mathcal{L}_a \\
 \downarrow \mathcal{I}_a & & & \downarrow \mathcal{I}_a & \\
 f & \leftarrow \in & \text{---} & M & \text{---} \subseteq \rightarrow D
 \end{array}$$

World models*

$$\begin{array}{ccccc}
 a & \leftarrow \in & \text{---} & \mathcal{T} & \text{---} \subseteq \rightarrow \mathcal{L}_a \\
 \downarrow \mathcal{I}_a & & & \uparrow \models & \\
 f & \leftarrow \in & \text{---} & M & \text{---} \subseteq \rightarrow D
 \end{array}$$

World logics*

World models allow to build world representations.
World logics allow to query them.

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Truth, Falsity and IDK

Definition (True assertion in a world logic). Let $L_W = \langle W, |=_W \rangle$ be a world logic with $W = \langle L_a, D, I_a \rangle$ its world model. Then, if $M |= a$, then we say that

a is True in M

Observation (Entailment checking = Question Answering Q/A). Checking $M |= a$ amounts to answering the question of whether a is true in M .

Observation (NOT knowing vs. knowing NOT). A positive answer to the above question means that a is **known to be true**. A negative answer **does NOT mean** that a is **known to be false**. It means that a is **NOT known to be true**, that is, **known to be IDK**.

Observation (Truth / Falsity, memory). The notions of Truth, Falsity and IDK (to be understood as “I don’t know”) as holding of an assertion are meaningful only if made with respect to a reference world logic.

Logical Entailment

Definition (Logical Entailment Relation). Let $L_W = \langle W, |=_W \rangle$ be a world logic with $W = \langle L_a, D, I_a \rangle$ its world model. Then, if a is True in M ($M |= a$) and M is a model of T , we write

$$T |= a$$

and say that the entailment relation $|=$, as used in $T |= a$, is a **Logical Entailment Relation**.

Observation (Logical Entailment Relation). Logical entailment relations allow us to model reasoning as the computation of which assertions, or sets of assertions follow from (that is, are logically entailed by) which assertions or set of assertions.

Definition (Axiom, Theorem). Let $L_W = \langle W, |=_W \rangle$ be a world logic with $W = \langle L_a, D, I_a \rangle$ its world model. With respect to a reference model M , with $M |= T$, if $T |= \underline{I}$ with $T, \underline{I} \subseteq L_a$, we call all the assertions in T , **axioms**, and all the assertions in $\underline{I}$ **theorems**.

Logical Entailment

Observation (Many-to-many world entailment relation). World entailment is a many-to-many relation. That is, depending on the language used, which usually extends the world model language, a theory may have multiple models, and vice versa. In which case the entailment relation will hold for all models and only for them.

Example (Many-to-many world entailment relation). A theory which contains the assertion «a person is a man **or** a woman» has a model where a person is a man and another model where a person is a woman.

Observation (Many-to-many logical entailment relation). World entailment is a many-to-many relation. That is, depending on the language used, which usually extends the world model language, a theory may entail multiple theories, and vice versa.

Observation (Many-to-many logical entailment relation). Logical entailment must be defined with reference to one or more reference models.

Logical entailment

Definition (Logical entailment) Let $\{M\}$ be a set of models of T with $T, \underline{I} \subseteq L$ and $a \in L$. Then we write

$$T \models_{\{M\}} \underline{I} \qquad (T \models_{\{M\}} a)$$

and say that T **logically entails** $\underline{I}$ (T **logically entails** a) with respect to $\{M\}$ if

$$\text{for all } M \in \{M\}, \text{ if } M \models T \text{ then } M \models \underline{I} \ (M \models a)$$

Observation (Logical entailment, reasoning). Logical entailment implements reasoning in the following way: “Whatever we derive from T must be true in the same sets of “relevant” models $\{M\}$ as T .”

Observation (Which set of models $\{M\}$). The choice of the models $\{M\}$ of T is arbitrary. It could be the only model of T , all models of T , a model selected because it is the intended model of T , and so on.

Notation ($\{M\}$). When no confusion arises we write $\models$ instead of $\models_{\{M\}}$.

Model and theory entailment

Terminology (Model entailment, theory entailment)). World entailment is also called **model entailment**. Logical entailment is also called **theory entailment**.

Observation (Model entailment, theory entailment)). Model entailment is a relation between a model and a theory. Theory entailment is a relation between two theories.

Observation (Using world entailment)). Model entailment allows to infer assertions from an input model, thus creating a linguistic representation of an analogical representation. This is the kind of reasoning process that in computer vision is performed when doing object or concept recognition.

Observation (Using theory entailment)). Theory entailment allows to infer assertions from an input theory, thus extending an existing linguistic representation of an analogic representation. This is the mainstream reasoning process used in logics.

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Properties of Logical Entailment

- Logical (theory) entailment links what one asserts as being the case with what is true in the model.
- Logical (theory) entailment is a formalization of our intuitive notion of reasoning.
- There are multiple notions of logical entailment, formalizing different notions of reasoning with wildly different properties.
- Still there are certain properties that we want all forms of logical entailment to satisfy

Logical entailment – properties (P1)

Definition (Reflexivity)

$$w \models w$$

Observation (Reflexivity). Every fact entails itself. Knowledge asserts itself as being knowledge. This is the essence of what knowledge is about. That is, if a formula is true in a representation (the memory of what we have perceived or been told), then I'll answer so any time I am asked.

Observation (Reflexivity). Reflexivity, to hold, requires that one is capable of recognizing two occurrences of the same formula (proposition, assertion, fact) to be the same. That is (as stated earlier on) we need a theory of (dis)similarity.

Observation (axioms, theorems). Because of reflexivity all axioms are theorems.

Logical entailment – properties (P2)

Definition (Cut)

If $T \models w_1$ and $\Sigma \cup \{w_1\} \models w_2$ then $T \cup \Sigma \models w_2$

Observation (Cut) There are two ways to interpret cut.

The first and most common relates to **efficiency**. That is, reasoning can be made efficient by dropping intermediate irrelevant results.

The second relates to **transitivity**. That is, reasoning can be computed by chaining independent reasoning sessions, something that people do all the time during their everyday life.

Logical entailment – properties (P3)

Definition (Compactness)

If $T \models w$ then there is a finite subset $T_0 \subseteq T$ such that $T_0 \models w$

Observation (Compactness). Compactness says that we do not need to use all we know to derive our goal.

This is a property extensively used by humans in common sense reasoning. We always minimize the amount of knowledge we use when reasoning.

This applies also in the case that what we know is infinite (which happen with infinite sets, for instance, with the natural numbers, or, more interestingly, with time and space coordinates and regions).

Logical entailment – properties (P4)

Definition (Monotonicity)

$$\text{If } T \models w \text{ then } T \cup \Sigma \models w$$

Observation (Monotonicity) Monotonicity implements a fundamental and intuitive property of knowledge, for instance of **scientific knowledge**. If knowledge increases then what can be derived from it via entailment can only increase.

At worst, the new knowledge will not change anything if the new piece of knowledge was implied by what is already known.

Logical entailment – properties (P5)

Definition (NonMonotonicity)

If $T \models w$ and $T \cup \Sigma$ then $\text{not} \models w$

Observation (NonMonotonicity) Monotonicity is a property which most often does not hold. This is extensively the case with common sense reasoning, a topic extensively studied in AI.

Historical AI example: the belief that all birds fly can be defeated by the fact that penguins are birds and they do not fly.

Historical scientific knowledge example: the discovery that it is the earth rotating around the sun, and not vice versa.

Everyday life example: perception in action is nonmonotonic, planning is nonmonotonic.

Practical point of view: the logics used in mathematical reasoning and in formal methods, as applied to, e.g., programming languages, are monotonic, while most logics defined in AI are nonmonotonic. Negation by failure, as implemented in relational DBs is nonmonotonic.

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Logic – expressiveness

Definition (World logic expressiveness). The expressiveness of a logic is formally defined by three parameters:

- The **world models** that is able to represent as part of the language
- The **expressiveness** of the language used to make assertions about the world model
- The set of **reasoning problems** that it allows to solve

Logic – expressiveness

Definition (World models). By world model we mean any of the seven world models we have formalized or any specialization which guarantees the properties of world model as defined before.

Example (Specialized world models). Examples:

- Domain languages;
- Concepts, etypes, entities, and corresponding percepts, as needed in health
- Concepts, etypes, entities as they are needed to describe the world as defined by the various types on LTLOs (e.g., domains, taska, applications)
- The intersection of two domain languages closed under formation rules

The intuition is that we must have full freedom in modeling the world we need to model, as long as we preserve the basic properties mentioned above and later for entailment (e.g. correctness and/ or completeness).

Logic – expressiveness

Definition (Language expressiveness). Given the ability of entailment to infer knowledge from what we already have in memory, the language must provide the ability to express the knowledge and the questions which are of interest

Example (Language expressiveness). Examples:

- Allowing to express generality /specificity (as in “women are people”);
- Allowing to express incomplete information (as in “This person is Fausto or Leonardo” (this information cannot be expressed in world models);
- Allowing to express negative information (as in “people are not animals”) (this information cannot be expressed in world models);

Logic – expressiveness

Definition (Reasoning). By **reasoning** we mean the process by which we derive new knowledge (in the case of world logics, a new assertion) from memory.

Definition (Reasoning problems). A **reasoning problem** is a [YES/NO] assertion question of the form “Does $M \models a$ hold?” or “Does $T \models a$ hold?”

Definition (Decision procedure). A **decision procedure** (for a specific reasoning problem) is an algorithm which provides an answer for a specific reasoning problem.

Example (Reasoning problems). Examples:

- Does the current memory have a model;
- Does an assertion follows for what is in memory
- Is an assertion consistent / inconsistent with the current memory;
- Do two assertions describe the same percept with respect to the current memory;
- Are two percepts disjoint with respect to the current meory.

Logic - expressiveness

Observation (A crucial trade-off). The more expressive the language, the more complex reasoning problems can be stated. In turn an increase in the complexity of the reasoning problem, results in an increase of the complexity class of the reasoning problem.

Terminology (Decidability of a Logic). A Logic is **decidable** if there exists a decision procedure which decides entailment in finite time, no matter the reasoning problem. A logic is **semi-decidable** if the decision procedure with returns YES or it does not terminate.

Example (Complexity of reasoning problems). Examples:

- Deciding whether two concepts in a CG (as formalized in LoC) are one the subsumption of the other is in the complexity class P;
- Deciding whether two etypes (entities) in a ETG (EG) (as formalized in LoD) where it is allowed to express partial knowledge, as in “A person is a male or a female”, is in the complexity class NP;

Logic – Correctness, completeness

Definition (Logic Correctness and completeness). We have five levels of correctness / completeness

- Language correctness/completeness: relates language to domain of interpretation
- Interpretation correctness/completeness: relates assertions to facts
- Theory correctness/completeness: relates theory to model, or, said in another way, theorems to true facts
- Entailment correctness/completeness: relates model to (a new) theory, or, said in another way, axioms to theorems / true facts
- Decision procedure correctness/completeness: relates theorems to computable theorems

Exercise (Logic Correctness and completeness). Provide formal definitions of the five notions of correctness / completeness provided above.

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World logics diversity

Observation (World logics diversity). We have the following

- **World model diversity.** Two world logics which are based on different world models are a priori not comparable. The only way to compare them is based on the translation mechanisms, introduced before, across world models
- **World entailment diversity.** The diversity of entailment of two logics based on the same world logic is in terms of the complex assertion used to define the entailment relation.

Family of world logics

Definition (Family of world logics). A family of world logics is a set of world logics which share possibly partially, the same world model.

Observation (World logics Family). The key idea behind a family of world logics is as follows:

- Select a so called **reference world model** $W = \langle L_a, D, I_a \rangle$.
- Define the so-called **base world logic**, that is the world logic $L_W = \langle W, |=_{L_a} \rangle$ implementing world entailment on assertions stating what is true in the reference world model
- Define a more complex world logic in a family from a simpler world logic as follows:
 - by extending the expressiveness of the language, to allow for new assertions, beyond assertions which just name the percepts and facts occurring in D . We call them **atomic** and **complex assertions** (denoting **percepts**, **facts/ perceptions**)
 - by extending the interpretation function, from atomic assertions (percepts, facts) to complex assertions (percepts, facts)
 - by extending accordingly the entailment relation, beyond world entailment to allow for assertional theories involving atomic as well as complex assertions.

More expressive world logics in the same family allow for more expressive reasoning.

World logics

- **LoE**: the **Logic of Entities**. It is the base world logic. It formalizes (only) the interpretation function based entailment of **world models**;
- **LoC**: the **Logic of concept definitions**. It formalizes how the **LoE language** can be extended to allow for **concepts**, whose meaning is defined by proper LoC concept **definitions**, and reasoning about them;
- **LoD**: the **Logic of etype descriptions**. It formalizes how the **LoE language** can be extended to allow for **etypes**, whose meaning is described by proper LoD etype **descriptions / definitions**, and reasoning about them;
- **LoCE**: LoE + LoC;
- **LoDE**: LoD + LoE
- **LoCD**: LoC + LoD
- **LoCDE**: LoE + LoC + LoD

Observation (world logics): We have seven world logics which map one-to-one to the seven world models seen above and that allow to reason about them. World Logic reasoning allows to make explicit all the knowledge which is implicit in the meaning relations implicitly encoded in concepts, etypes and entities.

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Key notions

- World Logics
- Truth, Falsity, IDK
- Canonical world entailment, world, model, logical and theory entailment
- Properties of logical entailment: reflexivity, cut, compactness, monotonicity, nonmonotonicity
- Reasoning as (model and theory) entailment, reasoning problem, decision procedure
- Properties of world logics: expressiveness, (semi)decidability, complexity, (five types of) correctness and completeness
- World Logic Diversity
- Base world logics: LoE, LoC, LoD
- Composed world logics: LoCD, LoCE, LoDE, LoCDE



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